

East Kitsap Riparian Assessment

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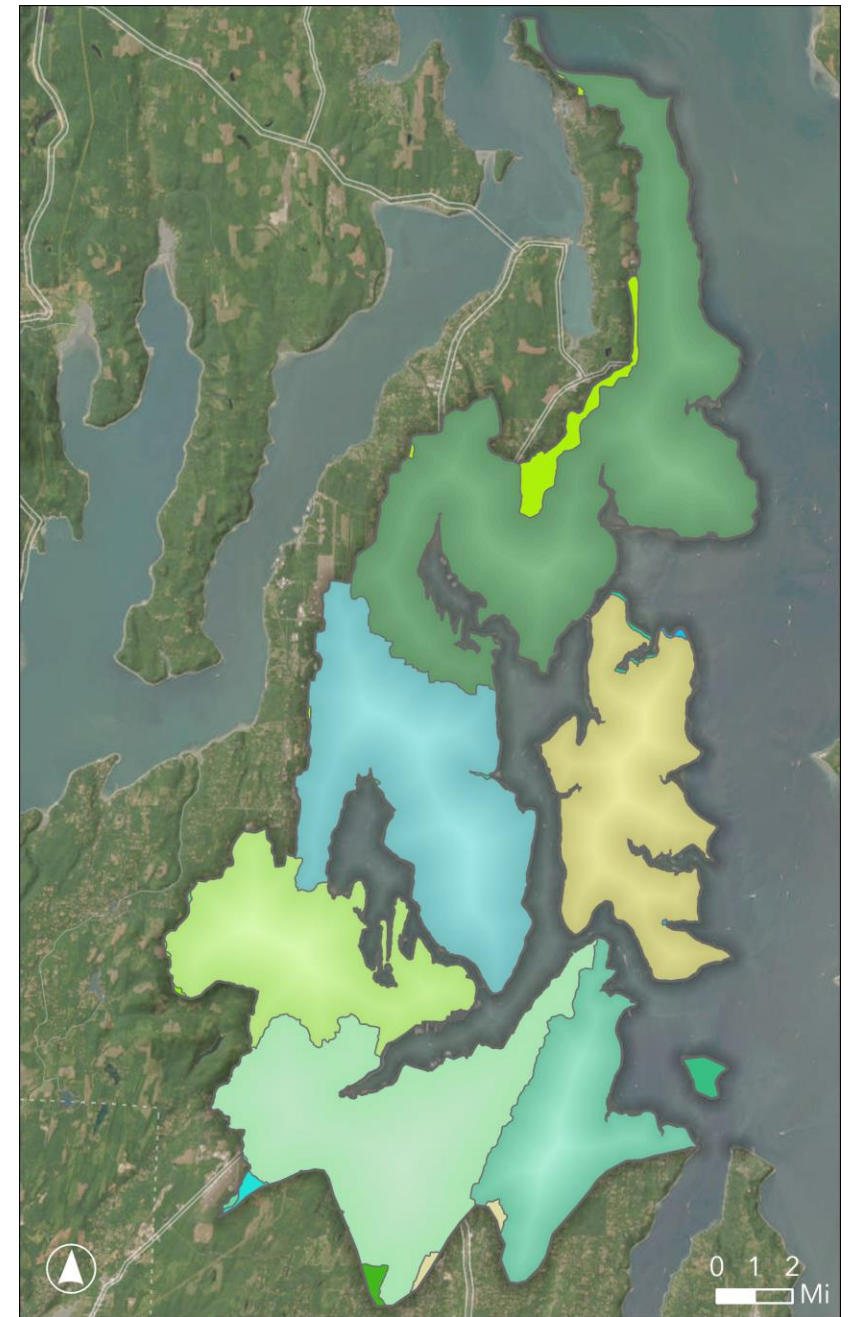
Riparian Assessment

- **Project Goal:** Data-driven/science-based approach to identifying and prioritizing riparian protection and restoration
- WWU Spatial Institute + Suquamish Indian Tribe
- Focus on East Kitsap streams and riparian zones



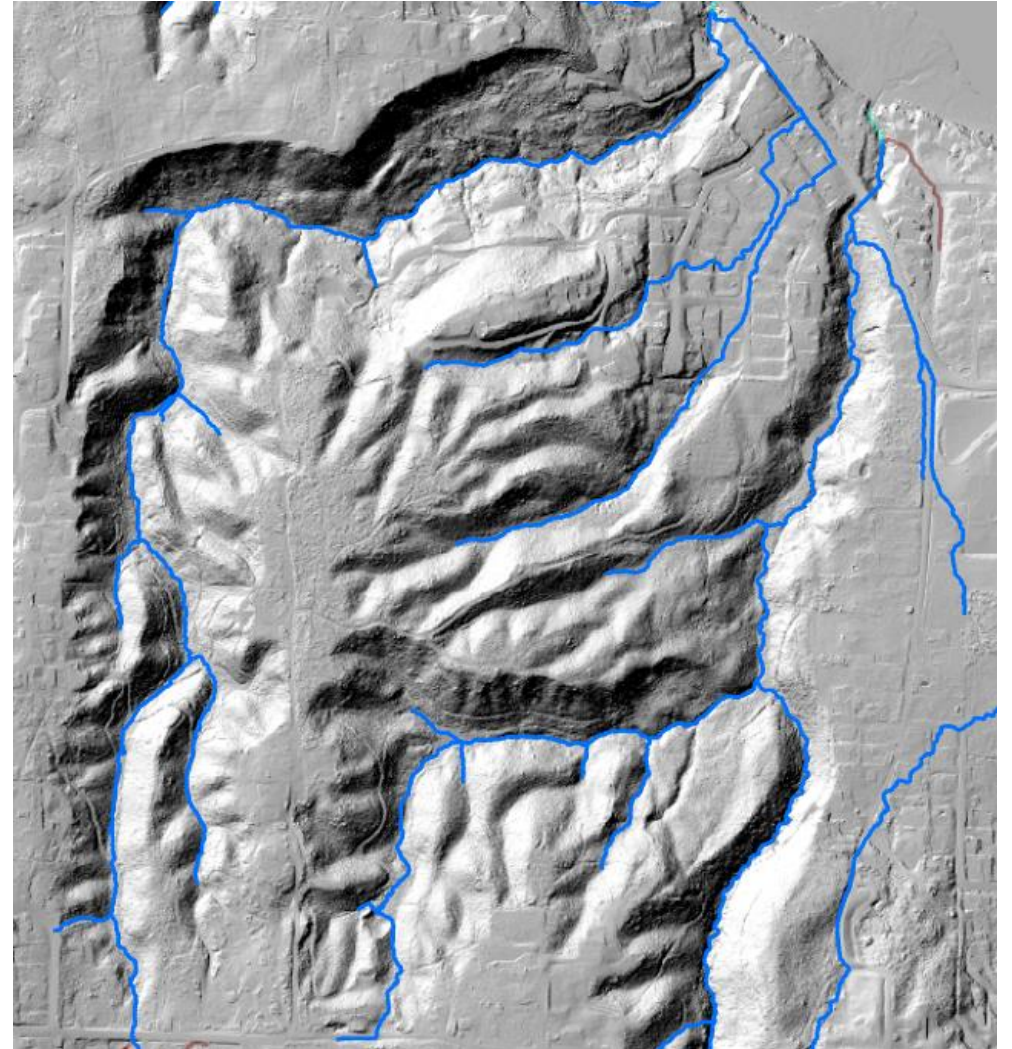
East Kitsap

- East Kitsap County watersheds that potentially support salmonid habitat
 - Not the entire county



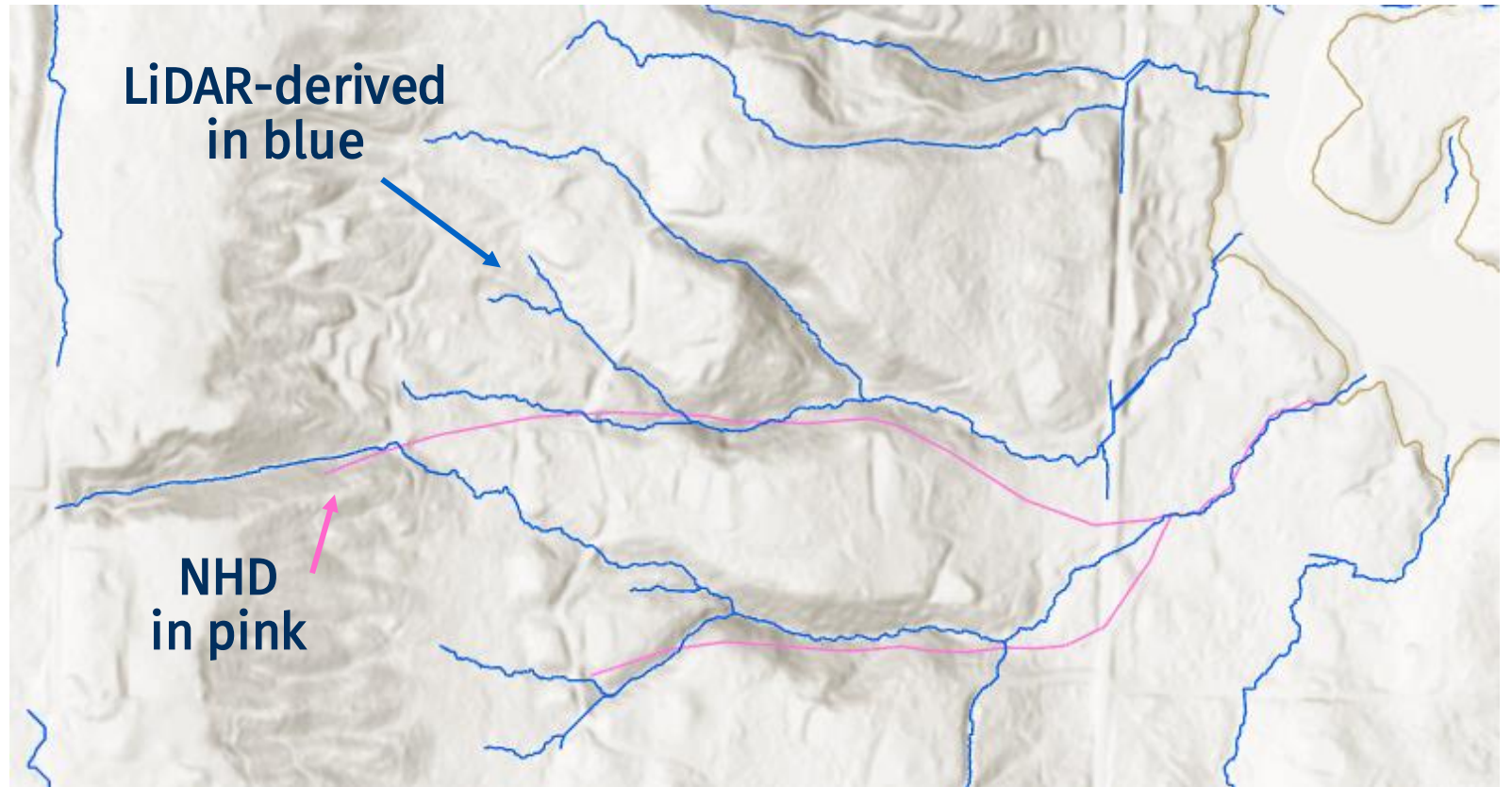
Riparian Assessment Data

- **Data Goal:** LiDAR-based stream network and riparian characterization focused on tree canopy height/shade and large wood inputs to streams
- Incorrect temperatures for salmonids affect egg development and rearing, species competition, disease susceptibility



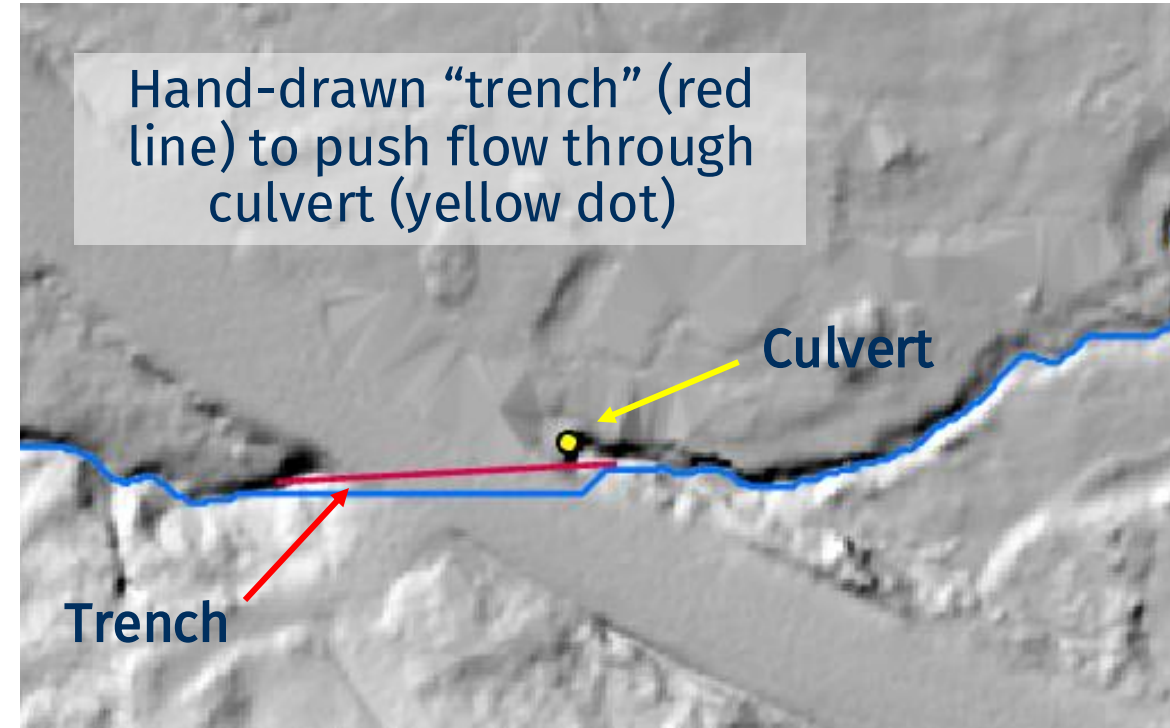
LiDAR-Derived Streams

- LiDAR allows us to update hydrography to a more locally useful spatial resolution



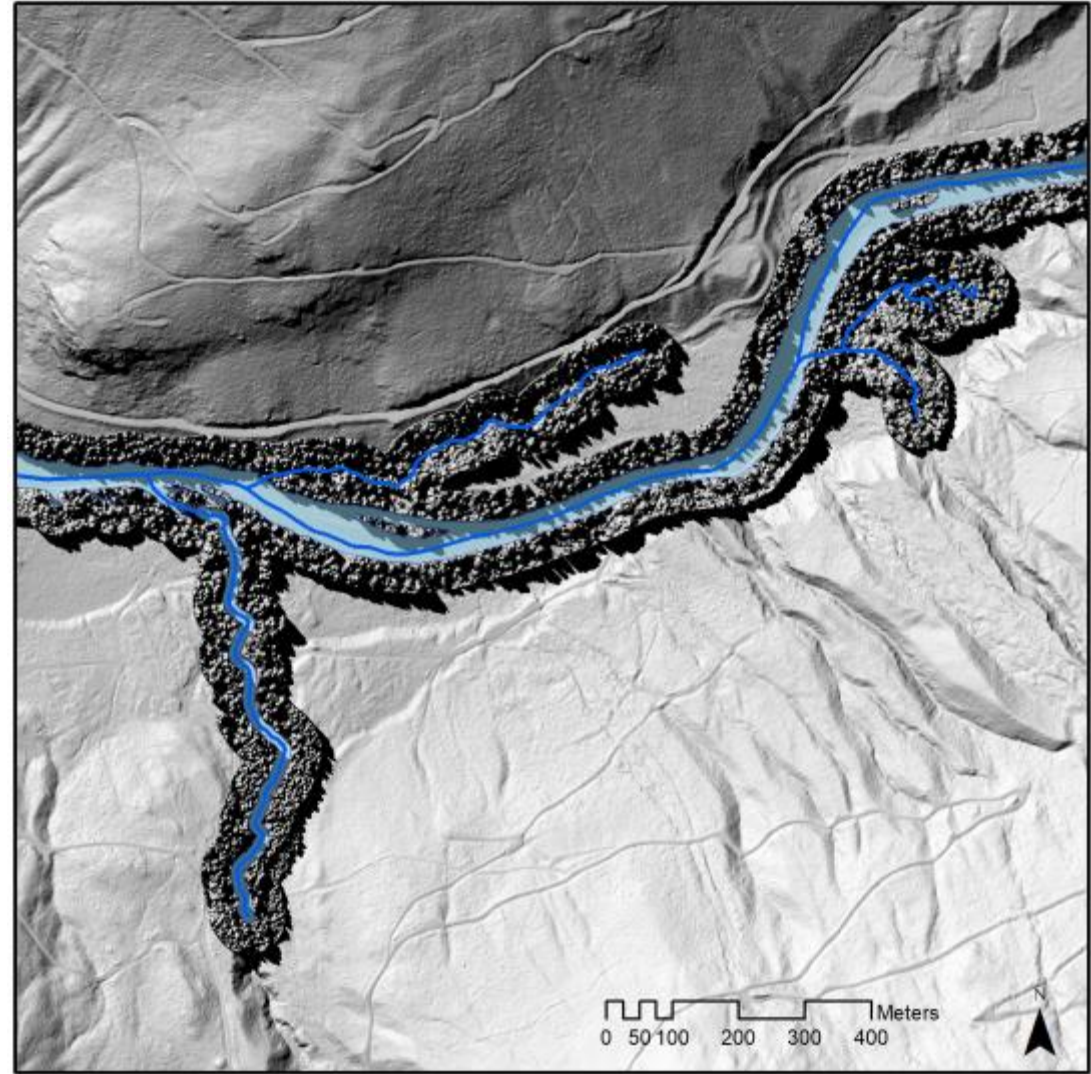
East Kitsap Streams

- Generation of single-thread hydrography using ArcGIS Pro & Python Scripting from Hyatt (2022)
 - Based on LiDAR
 - Focus on flow direction, sink filling, and flow accumulation
- Manual editing to direct flow through anthropogenic modifications like road culverts



Shade Modeling

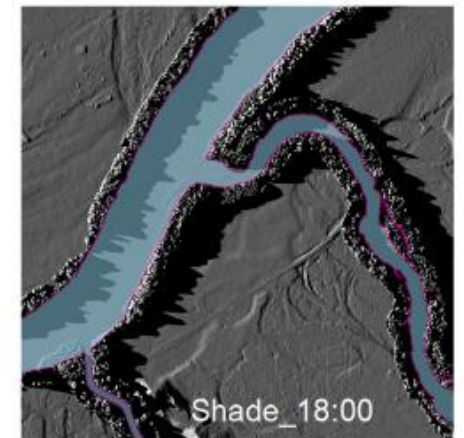
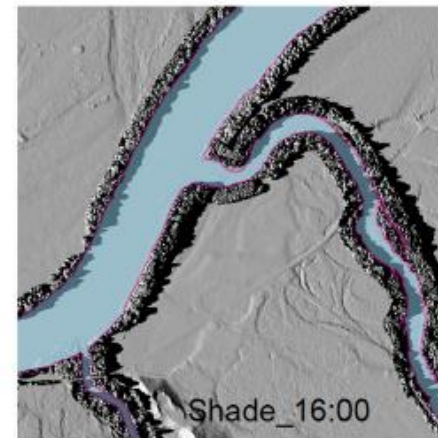
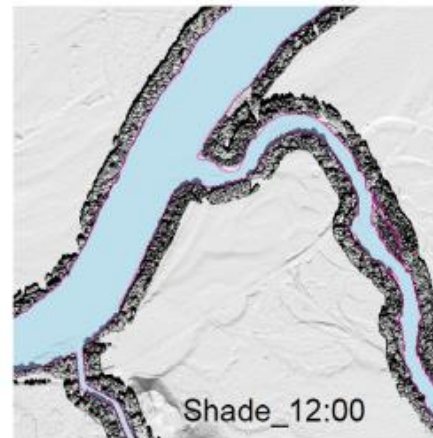
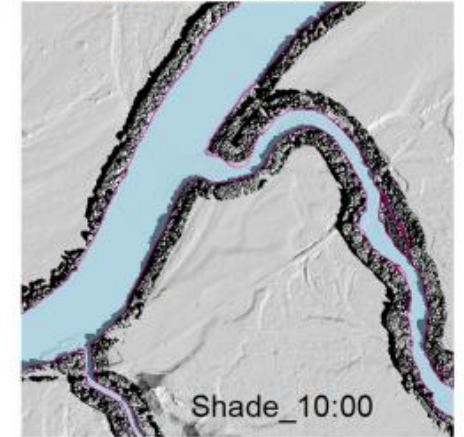
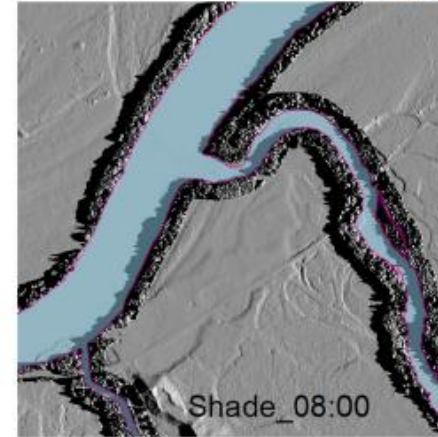
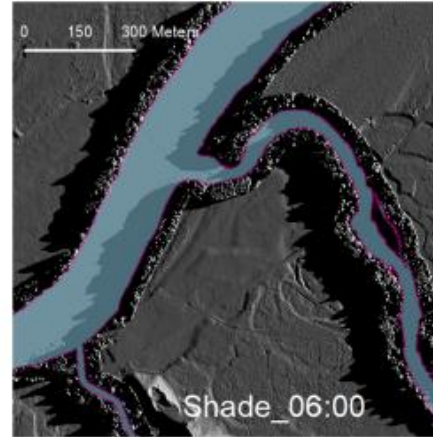
- Identify trees and canopy height in riparian zone buffer based on single-thread hydrography
- Model shade casting
- How long is each specific place in the sun?



Map from Hyatt (2022)

Final Takeaways

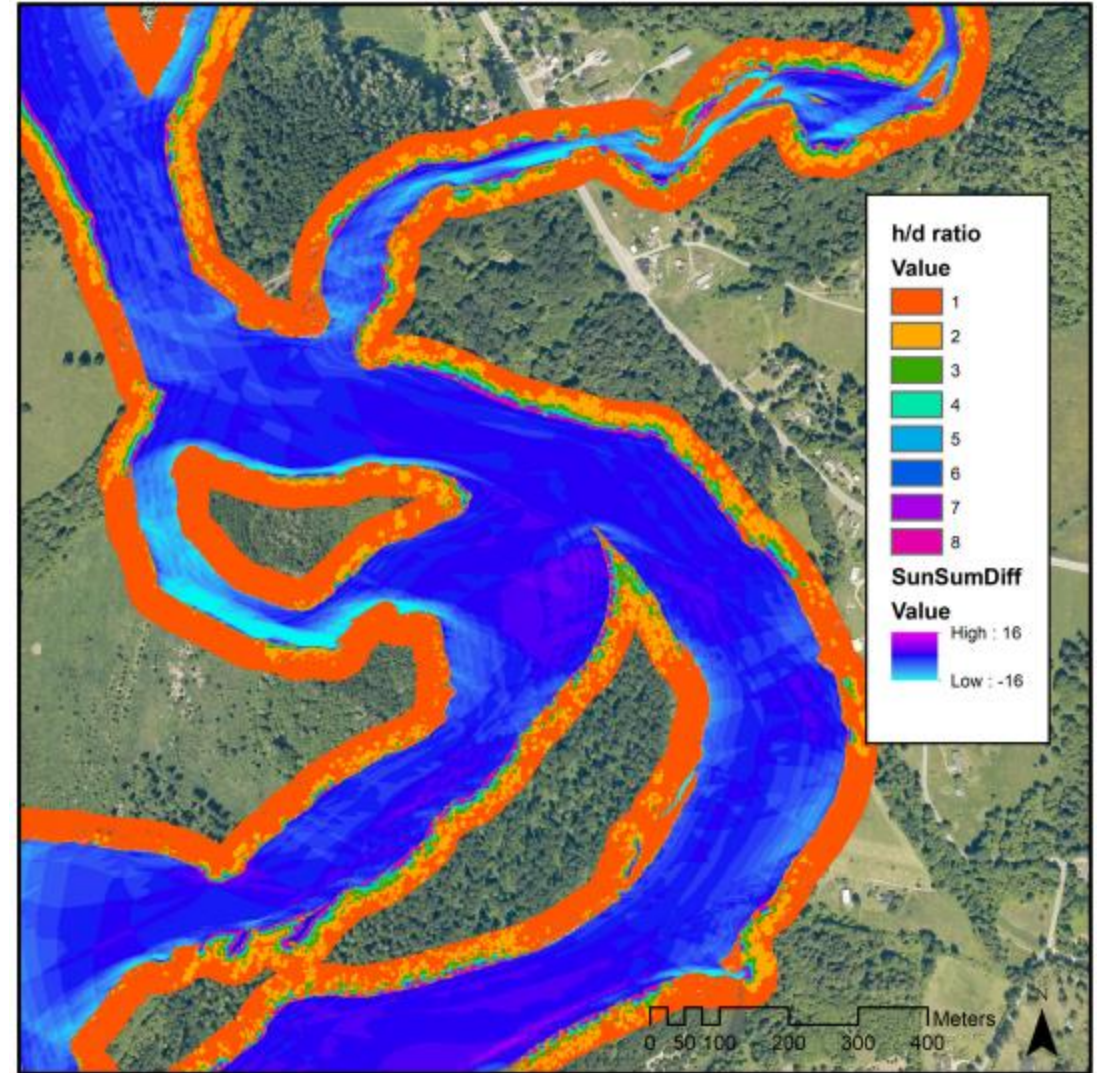
- Where are the degraded, fragmented riparian areas in East Kitsap watersheds?
- Where are the intact, functional riparian areas?



Map from Hyatt (2022)

Final Takeaways

- Highlight areas that are in greater need for riparian restoration to restore wood recruitment and stream shading
- Identify areas that are functioning and in need of long-term protection



Map from Hyatt (2022)

Thank you!

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